## Problem 1

(a) $\exists x.\, S(x) \wedge A(x)$ ✓

(b) $\forall x.\, T(x) \wedge A(x)$ ✗ $\forall x \in X:\ T(x) \wedge S(x) \Rightarrow A(x)$;

(c) $\exists! x\ T(x) \wedge \neg A(x)$ ✓

(d) $\exists x, y, z.\, T(x) \wedge \neg S(x) \ldots$ ✗

## Problem 2

(a)

| P | Q | R | ¬P | ¬Q | Q∧R | ¬(P∨(Q∧R)) (left) | ¬R | ¬Q∨¬R | (¬P)∧(¬Q∨¬R) (right) |
|---|---|---|---|---|---|---|---|---|---|
| T | T | T | F | F | T | F | F | F | F |
| T | T | F | F | F | F | F | T | T | F |
| T | F | T | F | T | F | F | F | T | F |
| T | F | F | F | T | F | F | T | T | F |
| F | T | T | T | F | T | F | F | F | F |
| F | T | F | T | F | F | T | T | T | T |
| F | F | T | T | T | F | T | F | T | T |
| F | F | F | T | T | F | T | T | T | T |

$\neg(P \vee (Q \wedge R)) = (\neg P) \wedge (\neg Q \vee \neg R)$ ✓

b)

(b) i will refer the table from (a)

| Q∨R | P∧(Q∨R) | ¬P∨(¬Q∨¬R) | ¬(P∧(Q∨R)) |
|---|---|---|---|
| T | T | F | F |
| T | T | T | F |
| T | T | T | F |
| F | F | T | T |
| T | F | T | T |
| T | F | T | T |
| T | F | T | T |
| F | F | T | T |

$\neg(P \wedge (Q \vee R)) \neq \neg P \vee (\neg Q \vee \neg R)$

Problem 3 : $P \text{ nand } Q = \neg(P \wedge Q)$ $P \text{ nand } P = \neg(P \wedge P) = \neg P$

(a) (i) $P \wedge Q = \neg(P \text{ nand } Q)$ ✓

(ii) $A \vee B = \neg(\neg A \wedge \neg B)$
$= \neg A \text{ nand } \neg B$
$= (A \text{ nand } A) \text{ nand } (B \text{ nand } B)$ ✓

(iii) $A \to B = \neg A \vee B$
$= \neg(A \wedge \neg B)$ ✓
$= A \text{ nand } \neg B$

(b) $\neg A = A \text{ nand } A$ ✓

(c) Always True $= T = A \vee \neg A = \neg(\neg A \wedge A)$
$= \neg A \text{ nand } A$
$= (A \text{ nand } A) \text{ nand } A$ ✓

Always False $= F = A \wedge \neg A$
$= \neg\neg(A \wedge \neg A)$
$= \neg(A \text{ nand } \neg A)$
$= \neg(A \text{ nand } (A \text{ nand } A))$
$= (A \text{ nand } (A \text{ nand } A)) \text{ nand } (A \text{ nand } (A \text{ nand } A))$ ✓

## Problem 4

1st Weighting (6vs6): Since all real coins weigh the same and the fake coin is lighter, Choose the higher plate since it included fake

2nd Weighting (3vs3): Split those 6 coins into two groups of 3 and compare the weight, Choose the higher (meaning lighter) plate since it included fake coin

3rd Weighting (1vs1): Choose random 2 coins from 3 coins and compare the weight

- if the scale is balances, the unweighted coin is fake
- if the scale tips, the lighter coin on the scale is fake

✓

## Problem 5

Contrapositive: "If $r^{\frac{1}{5}}$ is rational, then $r$ is rational"

$$r^{\frac{1}{5}} = \left(\frac{b}{a}\right)$$

$$\left(r^{\frac{1}{5}}\right)^5 = \left(\frac{b}{a}\right)^5$$

$$= \frac{b^2}{a^2} \leftarrow \text{rational}$$

✓

4 possible cases ① $x, y, w$ are even

$(2a)^2+(2b)^2+(2c)^2 = 4a^2+4b^2+4c^2 = 4(a^2+b^2+c^2)$

multiples of 4 are always even
therefore $z^2$ = even
only even × even = even therefore $z$ = even

② one of $x, y, w$ is odd

$(2a)^2+(2b)^2+(2c+1)^2 = 4a^2+4b^2+4c^2+4c+1$
$= 4(a^2+b^2+c^2+c)+1$

multiple of 4 are always even
even + 1 is odd $\therefore z^2$ = odd
only odd × odd = odd there $z$ = odd

③ two of $x, y, w$ are odd

$(2a)^2+(2b+1)^2+(2c+1)^2 = 4a^2+4b^2+4b+1+4c^2+4c+1$
$= 4a^2+4b^2+4b+4c^2+4c+2$
$= 4(a^2+b^2+b+c^2+c)+2$

the output is even
but if $z$ is even = $2i$
$z^2 = (2i)^2 = 4i^2$
$z^2$ is mulitle of 4 so it shoulod be dividsivle by 4
but $4(a^2+b^2+b+c^2+c)+2$ isn't divisible by 4 b.c +2
which contradict of $4i^2$ therefore output isn't even

④ all of $x, y, w$ are odd

$(2a+1)^2+(2b+1)^2+(2c+1)^2 = 4a^2+4a+1+4b^2+4b+1+4a^2+4c+1$
$= 4a^2+4a+4b^2+4b+4a^2+4c+3$
$= 4(a^2+a+b^2+b+c^2+c)+3$

same as ②

therefor only if $x, y, w$ are even $z$ can be even